OBJECTIVES OF LESSON 1:

1. INTRODUCTION TO AI
2. APPLICATIONS OF AI
3. HISTORY AND EVOLUTION
4. TYPES OF AI
5. DEEP LEARNING- FUNDAMENTAL CONCEPT
6. LINKS TO ARTICLES WILL BE PROVIDED FOR FURTHER RESEARCH
7. LINKS TO YOUTUBE VIDEOS WILL BE GIVEN
8. AI TOOLS TO BOOST STUDIES
9. CONCLUSION AND REMARKS

ARTIFICIAL INTELLIGENCE

ARTIFICIAL INTELLIGENCE (AI) IS A SET OF TECHNOLOGIES THAT ENABLE COMPUTERS TO PERFORM A VARIETY OF ADVANCED FUNCTIONS, INCLUDING THE ABILITY TO SEE, UNDERSTAND AND TRANSLATE SPOKEN AND WRITTEN LANGUAGE, ANALYZE DATA, MAKE RECOMMENDATIONS, AND MORE. SOURCE: GOOGLE CLOUD

APPLICATIONS OF AI

- **1. Healthcare**
- **Diagnostics**: AI systems analyze medical data to assist in diagnosing diseases (e.g., radiology, pathology).
- **Drug Discovery**: Identifying potential drug compounds and accelerating development.
- **Personalized Medicine**: Tailoring treatments to individual patients based on AI analysis of genetics and health data.
- **Healthcare Chat-bots**: Assisting patients with symptoms, scheduling appointments, and providing medical information.
- **Robotic Surgery**: AI-driven robots aid in minimally invasive surgeries with high precision

APPLICATIONS OF AI

- **2. Business**
- **Customer Service**: Chat-bots and virtual assistants handle customer queries efficiently.
- **Sales and Marketing**: Predictive analytics for lead scoring, customer segmentation, and targeted campaigns.
- **Fraud Detection**: AI identifies fraudulent activities in financial transactions.
- **Supply Chain Optimization**: Demand forecasting, inventory management, and logistics planning.
- **HR and Recruitment**: AI automates candidate screening and analyzes employee performance data.

HISTORY AND EVOLUTION OF AI

Who discovered it?	First innovation	Present
• John McCarthy coined the term "artificial intelligence" in 1956 and drove the development of the first AI programming language, LISP, in the 1960s. Early AI systems were rule-centric, which led to the development of more complex systems in the 1970s and 1980s, along with a boost in funding.	• **Kismet** • You can trace the research for Kismet, a "social robot" capable of identifying and simulating human emotions, back to 1997, but the project came to fruition in 2000. Created in MIT's Artificial Intelligence Laboratory and helmed by Dr. Cynthia Breazeal, Kismet contained sensors, a microphone, and programming that outlined "human emotion processes." All of this helped the robot read and mimic a range of feelings.	• 2023 was a milestone year in terms of generative AI. Not only did Open-AI release GPT-4, which again built on its predecessor's power, but Microsoft integrated Chat-GPT into its search engine Bing and Google released its GPT chat-bot Bard.

TYPES OF AI
BASED ON CAPABILITIES

- **Narrow AI (Artificial Narrow Intelligence - ANI)**
- **Definition**: AI designed to perform a single task or a limited range of tasks.
- **Characteristics**:
  - Highly specialized.
  - Cannot perform tasks outside its predefined scope.
- **Examples**:
  - Voice assistants like Siri or Alexa.
  - Recommendation systems (Netflix, Amazon).
  - Facial recognition systems.
  - Spam email filters.

- **General AI (Artificial General Intelligence - AGI)**
- **Definition**: AI that can perform any intellectual task that a human can do.
- **Characteristics**:
  - Capable of learning and reasoning across various domains.
  - Exhibits self-awareness and adaptability.
  - This is still theoretical and not yet achieved.
- **Potential Examples**:
  - A system capable of solving problems in different fields without specific training.

BASED ON FUNCTIONALITY

- **Reactive Machines**
- **Definition**: AI systems that operate solely based on current data without memory.
- **Characteristics**:
  - Cannot learn from past experiences.
  - Focused on specific tasks.
- **Examples**:
  - IBM’s Deep Blue (chess-playing AI).
  - Simple robotics with no adaptability.

BASED ON FUNCTIONALITY

- **Limited Memory**
- **Definition**: AI systems that can use past data for decision-making but have limited memory.
- **Characteristics**:
  - Learn from historical data to improve performance over time.
- **Examples**:
  - Self-driving cars (analyzing traffic patterns and past data).
  - Image recognition systems.

DEEP LEARNING – A FUNDAMENTAL CONCEPT

WHAT IS DEEP LEARNING?

IMAGINE YOUR BRAIN AS A GIANT MACHINE THAT LEARNS THINGS BY LOOKING AT EXAMPLES. FOR EXAMPLE:
IF YOU SEE A LOT OF PICTURES OF CATS, YOUR BRAIN LEARNS WHAT A CAT LOOKS LIKE.
LATER, IF SOMEONE SHOWS YOU A NEW PICTURE, YOU CAN TELL IF IT'S A CAT OR NOT.
DEEP LEARNING IS LIKE TEACHING A COMPUTER TO DO THE SAME THING! INSTEAD OF A BRAIN, IT USES SOMETHING CALLED A **NEURAL NETWORK**, WHICH IS A BUNCH OF CONNECTED "VIRTUAL BRAIN CELLS."

LINKS AND RESOURCES

- https://www.britannica.com/technology/artificial-intelligence
- https://youtu.be/5hNK7-N23eU?si=Awe-otjmFXKhBk9V
- https://youtu.be/Y46zXHvUB1s?si=pSXoMU2iyQO3P7JO

TOOLS YOU CAN USE

LEARNING RESOURCES

THESE RESOURCES TEACH AI CONCEPTS INTERACTIVELY:

AI FOR EVERYONE (COURSERA):

A BEGINNER-FRIENDLY COURSE BY ANDREW NG TO UNDERSTAND THE BASICS OF AI.

KHAN ACADEMY:

COVERS MATH AND PROGRAMMING BASICS, WHICH ARE ESSENTIAL FOR AI.

DEEPLIZARD YOUTUBE CHANNEL:

SIMPLE, VISUAL EXPLANATIONS OF MACHINE LEARNING AND NEURAL NETWORKS.

YOUTUBE - SIMPLILEARN AI TUTORIALS:

VIDEO TUTORIALS ON AI CONCEPTS WITH EXAMPLES AND CODE WALKTHROUGHS.